

MODULE 2

Linear regression

Linear regression is a type of supervised machine-learning algorithm that learns from the labelled datasets and maps the data points with most optimized linear functions which can be used for prediction on new datasets. It assumes that there is a linear relationship between the input and output, meaning the output changes at a constant rate as the input changes. This relationship is represented by a straight line.

The goal of linear regression is to find a straight line that minimizes the error (the difference) between the observed data points and the predicted values. This line helps us predict the dependent variable for new, unseen data.

Minimizing the Error: The Least Squares Method

To find the best-fit line, we use a method called least square. The idea behind this method is to minimize the sum of squared differences between the actual values (data points) and the predicted values from the line. These differences are called residuals.

The formula for residuals is:

$$Residual = y_i - \hat{y}_i$$

Where:

- y_i is the actual observed value
- $\hat{y}_i$ is the predicted value from the line for that x_i

The least squares method minimizes the sum of the squared residuals:

$$Sum of squared errors (SSE) = \sum (y_i - \hat{y}_i)^2$$

This method ensures that the line best represents the data where the sum of the squared differences between the predicted values and actual values is as small as possible..

Interpretation of the Best-Fit Line

- **Slope (m):** The slope of the best-fit line indicates how much the dependent variable (y) changes with each unit change in the independent variable (x). For example if the slope is 5, it means that for every 1-unit increase in x, the value of y increases by 5 units.
- **Intercept (b):** The intercept represents the predicted value of y when x = 0. It's the point where the line crosses the y-axis.

Multiple Linear Regression

Multiple linear regression involves more than one independent variable and one dependent variable. The equation for multiple linear regression is:

$$\hat{y} = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_n x_n$$

where:

- $\hat{y}$ is the predicted value
- $x_1, x_2, \dots, x_n$ are the independent variables
- $\theta_1, \theta_2, \dots, \theta_n$ are the coefficients (weights) corresponding to each predictor.
- θ_0 is the intercept.

The goal of the algorithm is to find the best Fit Line equation that can predict the values based on the independent variables.

In regression set of records are present with X and Y values and these values are used to learn a function so if you want to predict Y from an unknown X this learned function can be used. In regression we have to find the value of Y, So, a function is required that predicts continuous Y in the case of regression given X as independent features.

Gradient Descent for Linear Regression

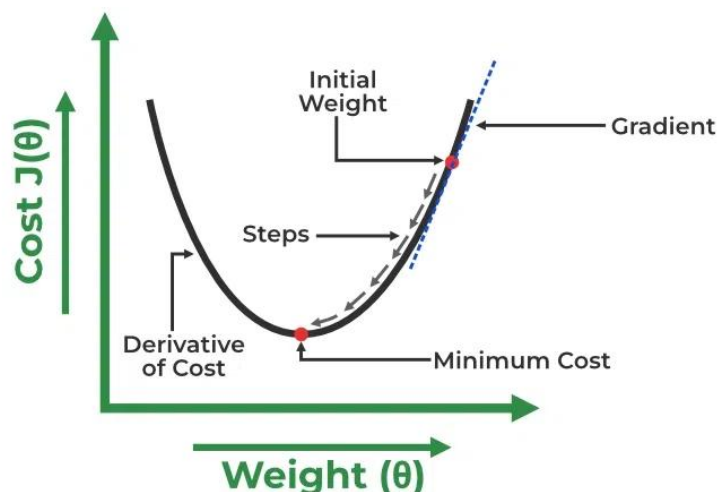
Gradient descent is an optimization technique used to train a linear regression model by minimizing the prediction error. It works by starting with random model parameters and repeatedly adjusting them to reduce the difference between predicted and actual values.

Gradient Descent

How it works:

- Start with random values for slope and intercept.
- Calculate the error between predicted and actual values.
- Find how much each parameter contributes to the error (gradient).
- Update the parameters in the direction that reduces the error.
- Repeat until the error is as small as possible.

This helps the model find the best-fit line for the data.



Gradient Descent is an iterative optimization algorithm used to minimize a cost function by adjusting model parameters in the direction of the steepest descent of the function's gradient. In simple terms, it finds the optimal values of weights and biases by gradually reducing the error between predicted and actual outputs.

Learning rate is an important hyperparameter in gradient descent that controls how big or small the steps should be when going downwards in gradient for updating model parameters. It is essential to determine how quickly or slowly the algorithm converges toward minimum of cost function.

1. If Learning rate is too small: The algorithm will take tiny steps during iteration and converge very slowly. This can significantly increase training time and computational cost especially for large datasets.

2. If Learning rate is too big: The algorithm may take huge steps leading to overshooting the minimum of cost function without settling. It fails to converge causing the algorithm to oscillate. This process is termed as exploding gradient problem.

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems. Unlike linear regression which predicts continuous values it predicts the probability that an input belongs to a specific class. It is used for binary classification where the output can be one of two possible categories such as Yes/No, True/False or 0/1. It uses sigmoid function to convert inputs into a probability value between 0 and 1.

Types of Logistic Regression

Logistic regression can be classified into three main types based on the nature of the dependent variable:

- 1. Binomial Logistic Regression:** This type is used when the dependent variable has only two possible categories. Examples include Yes/No, Pass/Fail or 0/1. It is the most common form of logistic regression and is used for binary classification problems.
- 2. Multinomial Logistic Regression:** This is used when the dependent variable has three or more possible categories that are not ordered. For example, classifying animals into categories like "cat," "dog" or "sheep." It extends the binary logistic regression to handle multiple classes.
- 3. Ordinal Logistic Regression:** This type applies when the dependent variable has three or more categories with a natural order or ranking. Examples include ratings like "low," "medium" and "high." It takes the order of the categories into account when modeling.

Naive Bayes

Naive Bayes is a machine learning classification algorithm that predicts the category of a data point using probability. It assumes that all features are independent of each other. The main idea behind the Naive Bayes classifier is to use Bayes' Theorem to classify data based on the probabilities of different classes given the features of the data.

Decision tree

A decision tree is a supervised learning algorithm used for both classification and regression tasks. It has a hierarchical tree structure which consists of a root node, branches, internal nodes and leaf nodes. It works like a flowchart help to make decisions step by step where:

- Internal nodes represent attribute tests
- Branches represent attribute values
- Leaf nodes represent final decisions or predictions.

So it's literally a *tree* where:

- **Internal nodes** = questions (conditions on features)
- **Branches** = possible answers (yes/no or ranges)
- **Leaves** = final decision or class label

How to split

1. Entropy (used in ID3)

$$Entropy = -\sum p_i \log_2(p_i)$$

→ Measures disorder. 0 = pure, 1 = mixed.

2. Information Gain

$$Gain = Entropy(\text{parent}) - \text{Weighted average of children entropies}$$

→ The higher the gain, the better the split.

ID3 Algorithm (Iterative Dichotomiser 3)

You'll probably need this one for your test.

It works as:

1. Calculate **entropy** of the target (class)
2. For each feature:
 - Calculate **information gain** for all possible splits

3. Choose the feature with **maximum information gain** to split
4. Repeat this process **recursively** for each branch until:
 - All records in a branch belong to the same class
 - Or no features remain

Advantages and disadvantages

- ✓ Very easy to understand and visualize
- ✓ Works for both numerical and categorical data
- ✓ No need for feature scaling
- ✗ Can **overfit** (memorize training data)
- ✗ Small changes in data can change the tree

Linear v/s logistic Regression

Linear Regression	Logistic Regression
Linear Regression is used to handle regression problems	Logistic regression is used to handle the classification problems.
In Linear Regression, we predict a continuous value	In Logistic Regression, we predict a categorical value
Here no activation function is used.	Here activation function is used to convert a linear regression equation to the logistic regression equation
no threshold value is needed.	Here a threshold value is added.
Forms a straight line	Forms an S-shaped (sigmoid) curve